

Total number of printed pages-5

1-23 (PHY1104C)

2023

PHYSICS

(Core Course)

Paper : PHY1104C

(Mechanics and Properties of Matter)

Full Marks : 45

Time : Two hours

**The figures in the margin indicate
full marks for the questions.**

Section-I

(Laws of Motion)

1. Answer **any two** from the following questions : $3 \times 2 = 6$

(a) State Newton's 2nd law of motion.
Hence establish the law of conservation
of linear momentum. $1 + 2 = 3$

(b) Find the position of centre of mass of
a semi-circular arc.

Contd.

(c) Define work done. A force $\vec{F} = 2\hat{i} - \hat{j} + 3\hat{k}$ newton acting on a body displaces it from its initial position of $\hat{i} - \hat{j} + 2\hat{k}$ to $3\hat{i} + 2\hat{j} - \hat{k}$. Calculate the work done by the force. 1+2=3

(d) Show that in a centre of mass frame, the total linear momentum of a system is always zero.

2. Answer **any one** of the following questions: 5

(a) Show that total angular momentum of a system is equal to the sum of its orbital angular momentum and spin angular momentum.

(b) Derive an expression for moment of inertia of a solid sphere about its own diameter.

Section-II

(Gravitation)

3. Answer **any two** from the following questions: 3×2=6

(a) Define inertial and gravitational mass. Are they same? 2+1=3

- (b) Express graphically the variation of gravitational potential and intensity with distance due to a solid sphere.
- (c) Prove that the angular momentum of a system under the influence of central force field always remains constant.
- (d) State Kepler's laws on planetary motion.

4. Answer **any one** of the following questions : 5

- (a) Explain how a two-body problem involving central forces, can always be reduced to the form of a one-body problem. What is reduced mass? 4+1=5

- (b) Find an expression for gravitational potential due to a spherical shell at a point inside the shell.

Section-III

(Elasticity)

5. Answer **any two** from the following questions : 3×2=6

- (a) Draw the stress-strain diagram. Locate the elastic limit and yield point in the diagram. 2+1=3

- (b) In case of an elongation produced in a cylindrical rod of length l_0 and volume V_0 due to the application of a pair of stretching force, prove that

$$\frac{V}{V_0} = \frac{1 + \epsilon}{(1 + \sigma\epsilon)^2}$$

where σ = Poisson's Ratio, ϵ = longitudinal strain.

- (c) Find the time period of oscillation for a torsional pendulum.
- (d) Derive an expression for strain energy gained by a deformed elastic body.

6. Answer **any one** from the following questions : 5

- (a) Derive the expression for torsional rigidity for a uniform cylindrical solid rod.

(b) Prove that

(i) $Y = 3K(1 - 2\sigma)$

(ii) $Y = 2\eta(1 + \sigma)$

3+2=5

Section-IV
(Fluid Mechanics)

Answer **any two** from the following
questions : $3 \times 2 = 6$

- (a) What do you mean by Newtonian hypothesis for viscous force in liquid flow? Define 1 poise.
- (b) Find the continuity equation for liquid flow.
- (c) The speed of air going below the wings of a helicopter is 100 m/s . What will be the speed of air going above the wings to produce an uplift force of 1000 N/m^2 ?

8. Answer **any one** from the following
questions : 6

- (a) Derive Poiseuille's equation for liquid flow through a horizontal capillary tube.
- (b) Derive Bernoulli's theorem for liquid flow. How does a bowler apply the theorem to give a proper swing to the cricket ball? $4 + 2 = 6$